AS1: independent initial lane loads and the existing forward model

Elementary VSL lane-drop task review

19 October 2026

Status. Bounded AS1 review completed; exact amendment submitted at its user checkpoint. Step 3 remains open. AS2's observation contract and AS3's action/gradient conclusions are not established here.

Traffic meaning. The same ending-lane population can face a crowded adjacent lane or an uncrowded adjacent lane even when the continuing total is identical. Allowing that distinction requires independent initial lane loads, not a different traffic mechanism. The existing H1 preparation, finite T1 terminal, S-E-P allocation, R-E source, O-R renewal and W-F work laws support the asymmetric family below under their retained sufficient bounds. This is a conditional mathematical qualification, not empirical admission.

1 The exact initialization amendment

E is the ending lane, A its adjacent continuing lane and B the far continuing lane. The same uniform longitudinal footprint $[0, \ell]$ occupies the preparation reach at the decision, with $0 < \ell \le L_p$; the terminal of length $b > 0$ and the recovery account are empty. Set $L = L_p + b$ and

$$e_0, a_0, b_0 \ge 0, \qquad r = e_0 + a_0, \qquad \rho_0 = e_0 + a_0 + b_0 = r + b_0, \qquad M = \ell\rho_0 > 0. \tag{1}$$

There is *no* requirement $a_0 = b_0$. Equal loading remains an admitted special case. Longitudinal homogeneity within each lane is retained. There are no postdecision external arrivals. One common constant advice $v \in [v_-, v_+]$, $0 < v_- < v_+$, is applied to each fixed true initialization.

Define the same fundamental diagram and inverse boundary mark:

$$\begin{gathered} K = wk_j, \quad k_c(v) = \frac{K}{w+v}, \quad C(v) = \frac{Kv}{w+v}, \\ c(u) = \frac{Ku}{w+u}, \qquad U(g) = \frac{wg}{K-g}. \end{gathered} \tag{2}$$

Retain all existing sufficient restrictions, now with the actual three-lane mass in (1):

$$\begin{aligned} &w, k_j, h, a_r > 0, \quad \gamma \ge 1, \quad e_0, a_0, b_0 \le k_c(v_+), \\ &v_+(e_0 + a_0) < R \le C(v_-), \quad v_+ b_0 \le R_B \le C(v_-), \quad M < k_j b. \end{aligned} \tag{3}$$

The fixed physical coefficient and root condition are unchanged:

$$\begin{gathered} c_0 = 1 + \delta, \quad \delta, \beta \ge 0, \quad \chi = \frac{\beta}{C_\star v_\star}, \quad v_\star = v_+, \quad C_\star = C(v_\star), \\ \eta = c_0 - \beta(1 + v_+/w) > 0, \qquad c_{\max} = c_0 + \beta. \end{gathered} \tag{4}$$

The anchor defining χ stays fixed across advice; $m = 1$ veh-eq is the same physical mass normalization. All geometry and physical coefficients are shared across the contrasting initial states. The initial-dependent r is derived true state, not automatically a shared observer input. Density has units veh-eq/length, rates have units veh-eq/time, and M has units veh-eq; b_0 is a density and b a length.

2 Preparation, handoff and storage: no lane equality is used

The retained rule is

$$\begin{gathered} S(v,a)=\min\{C(v),w(k_j-a)\}, \qquad f=\min\{ve,(S(v,a)-va)/\gamma\}, \\ \dot e=-f/h, \quad \dot a=f/h, \quad \dot b_0=0. \end{gathered} \tag{5}$$

The transfer cancels in $e+a=r$. For every admitted advice,

$$r<R/v_+\le R/v\le C(v)/v=k_c(v). \tag{6}$$

Consequently $0\le e\le e_0$, $a=r-e<k_c(v)$ and $S=C(v)$. The scalar right-hand side is Lipschitz and nonnegative toward the invariant boundaries. No equality with B is needed. With progress $z=vt$ and $\Delta(v)=k_c(v)-r>0$,

$$e_z=-\frac{1}{h}\min\{e,(\Delta+e)/\gamma\}, \qquad a=r-e. \tag{7}$$

The two explicit branches remain

$$e=e_se^{-(z-z_s)/h} \quad \text{or} \quad e=(e_s+\Delta)e^{-(z-z_s)/(\gamma h)}-\Delta, \tag{8}$$

with a receiving-to-donor switch at $e=\Delta/(\gamma-1)$ for $\gamma>1$. For $\gamma=1$ the donor branch is active throughout.

Let $t_f^p=(L_p-\ell)/v$, $t_r^p=L_p/v$. During that pulse only,

$$\dot F_E=ve(t), \quad \dot F_A=v(r-e(t)), \quad \dot F_B=vb_0. \tag{9}$$

Let $s(t)=\ell$ before t_f^p, $s=L_p-vt$ during the pulse, and $s=0$ after t_r^p. With $J_{\mathrm{lat}}=\int_0^t sf/h\,\mathrm{d}u$, the balances are

$$\begin{aligned} se+F_E+J_{\mathrm{lat}}&=\ell e_0, & sa+F_A-J_{\mathrm{lat}}&=\ell a_0, \\ sb_0+F_B&=\ell b_0, & s\rho_0+F_E+F_A+F_B&=M. \end{aligned} \tag{10}$$

Each identity follows from its initial value and its derivative before, during and after the pulse. Early-merged mass enters A once and is not later counted as forced E service.

For $j\in\{E,A,B\}$, retain

$$P_j(t)=F_j(t-b/v), \quad N_j=F_j-Y_j, \quad q_j=P_j-Y_j, \quad \lambda_j=\dot P_j, \quad g_j=\dot Y_j, \tag{11}$$

with zero negative-time histories. N_j is physical terminal stock; q_j is eligible waiting mass. The far lane has $Y_B=P_B$ since $vb_0\le R_B$. Its density is not derived from A.

The terminal entrance vacancy is

$$V_j^{\mathrm{vac}}(t)=\begin{cases} k_jwt-F_j(t), & 0<t\le b/w, \\ Y_j(t-b/w)+k_jb-F_j(t), & t>b/w. \end{cases} \tag{12}$$

All entry rates are at most $C(v)<wk_j$. The early expression is positive; the later expression is at least $k_jb-M>0$. Thus finite receiving accepts the preparation handoff under the retained conservative total-mass bound. This protects the homogeneous upstream reduction; it is not a proof of general spillback dynamics outside this sufficient domain.

The eligible pulse spans $t_f=(L-\ell)/v$ to $t_r=L/v$. Inside it,

$$\lambda_A+\lambda_E=vr<R, \qquad \lambda_A'=-\lambda_E'\in[0,L_A], \qquad L_A=v_+^2e_0/h. \tag{13}$$

E offers decrease and A offers increase. These facts use $e+a=r$ and the donor upper bound $f\le ve$, not $a_0=b_0$.

3 The selected point service and recovery are unchanged

Free marks are $u_j = v$ and queued marks $u_j = U(g_j)$. Retain

$$c_E = c_0 + \chi g_A [u_A - u_E]_+, \qquad H(g; a, u) = g\{c_0 + \chi a[u - U(g)]_+\}. \tag{14}$$

Here the argument a in H denotes an A *flow*, not the upstream density. On the selected feasible interval, $H_g \geq \eta > 0$ and $x/c_{\max} \leq H^{-1}(x; a, u) \leq x/c_0$. The proof uses bounds on flows, marks and fixed coefficients, not initial lane equality.

The S-E-P branches are retained exactly:

Current mode	Service and free-feasibility condition
A/E free	$g_A = \lambda_A$, $g_E = \lambda_E$ if $\lambda_A + c_0\lambda_E \leq \mathcal{B}$.
A free, E queued	$g_A = \lambda_A \leq \mathcal{B}$, $H(g_E; \lambda_A, v) = \mathcal{B} - \lambda_A$.
A/E queued	$g_A = \mathcal{B}/2$, $H(g_E; \mathcal{B}/2, U(\mathcal{B}/2)) = \mathcal{B}/2$.
A queued, E free	$g_E = \lambda_E$, $g_A = \mathcal{B} - c_0\lambda_E$ if $c_0\lambda_E \leq \mathcal{B}/2$; otherwise E queues.

Retain strict-shortfall queue birth, actual-zero free-admission retesting, maximal feasible free continuation at equality, and right-hand data at arrival jumps. There is no new priority, timer, smoothing or branch latch.

For E's insertion mark u, retain

$$\begin{aligned} V(\theta; u) &= \sqrt{(w+u)^2 + 2wa_r\theta} - w, \\ D(\theta; u) &= [R - c(V(\theta; u))]_+, \\ p(t) &= \frac{1}{m}\int_{[Y_E(t)-m]_+}^{Y_E(t)} D(t - T(\xi); u_E(T(\xi)))\,\mathrm{d}\xi, \qquad \mathcal{B} = [R - p]_+. \end{aligned} \tag{15}$$

T is E's mass-rank admission-time inverse. Only E deposits loss. Recent mass replaces oldest retained weight; profiles also age. The account retains history, not just its current reading, and it does not keep the strictest prior cap. V is an equivalent boundary-speed quantity, not a vehicle trajectory. Neither the overlap approximation nor its physical validity is strengthened by this initialization amendment.

4 Forward qualification carries to every admitted asymmetric state

Proposition 1. *For* (1)–(4)*, the unchanged laws and causal guards have the same conditional unique complete forward continuation for every advice in the original interval. Mass, feasible service and finite terminal storage are respected. All traffic clears in finite time.*

Proof. The preceding preparation/storage argument supplies the same bounded, monotone eligible pulse as the accepted W-F proof. To make the remaining dependencies explicit, put $G = R/c_0$ and retain its bounds

$$p \leq p_{\max} := \frac{G}{m}\int_0^{m/G} D(\theta; 0)\,\mathrm{d}\theta < R, \quad B_{\min} = R - p_{\max} > 0, \quad |\dot{\mathcal{B}}| \leq L_B := 2GR/m + a_r k_j. \tag{16}$$

The account bound fills the youngest possible ages at the maximal admitted E rate. Strictness uses $a_r > 0$, not an imposed service floor. The history comparison for identical past and two subsequent flow/mark histories is

$$|\Delta p(t)| \leq \frac{1}{m}\left(5R\int_{t_0}^{t} |\Delta g_E|\,\mathrm{d}s + k_j G\int_{t_0}^{t} |\Delta u_E|\,\mathrm{d}s\right). \tag{17}$$

Together with $H_g \geq \eta$, this gives the same short-interval contraction on each valid mode. It does not assume a positive old E rate or initial symmetry. No numerical timestep is being defined.

At the front A's ordinary offer is below R. If joint free work fits, free E has $v \geq U(R)$ and produces no recovery loss. Joint work $vr+\delta\lambda_E$ decreases, so both remain free. Otherwise E's strict shortfall gives a compatible queue birth. At any later E clearance with free A, $\lambda_A + c_0\lambda_E \leq \mathcal{B}$; with queued A, $c_0\lambda_E \leq \mathcal{B}/2$. Free E then adds no loss, old loss ages/retires, and its offer decreases. These tests persist through any A clearance. After the rear there are no arrivals. E has at most one waiting episode under these specific assumptions, not under finite-cohort initialization alone.

While E queues, A birth is at $\lambda_A = \mathcal{B}$, with right-hand queue growth $\mathcal{B}/2 \geq B_{\min}/2$. A clearance requires $\lambda_A \leq \mathcal{B}/2$, leaving a free margin at least $B_{\min}/2$. The bounds (13) and (16) give a common positive separation for repeated A births/clearances,

$$\tau_0 = \frac{B_{\min}}{2(L_B + 2L_A)} > 0. \tag{18}$$

E clearance and the front/rear add only finitely many exceptions. The same feasible guards therefore join the unique local histories without event accumulation. After the rear, whenever $Q = q_A + q_E > 0$, $g_A + g_E \geq B_{\min}/c_{\max}$. Hence

$$T_{\text{clear}}(v) \leq L/v + c_{\max}M/B_{\min}, \quad T_{\max} = L/v_- + c_{\max}M/B_{\min} \tag{19}$$

is a common sufficient horizon. Every inequality depends on r, separate B admission, total mass or common coefficients. None requires $a_0 = b_0$. This transfers the accepted conditional forward argument; it does not establish smooth advice dependence at all queue boundaries. □

The complete balances and local total system travel time are consequently

$$s\rho_0 + \sum_j N_j + \sum_j Y_j = M,$$
$$J(v) = \int_0^{T_{\max}} \left(M - \sum_j Y_j(t) \right) \mathrm{d}t = \frac{M(L - \ell/2)}{v} + \int_0^{\infty} (q_A + q_E)\,\mathrm{d}t. \tag{20}$$

The first term integrates each initial position's remaining distance $(L - x)$ over all three lanes. It needs $\rho_0 = e_0 + a_0 + b_0$, not symmetry. B has no eligible queue in this domain. Completion remains at the choke; retained effects of completed E traffic affect unfinished followers only.

5 A nonempty fixed-E, fixed-continuing-total asymmetric family

In general, hold $e_0 = e$ and $a_0 + b_0 = c$ fixed and set

$$a_0 = c/2 + d, \qquad b_0 = c/2 - d, \qquad r = e + c/2 + d, \quad M = \ell(e + c). \tag{21}$$

The allowed d obeys the existing bounds

$$\begin{gathered} -c/2 \leq d \leq c/2, \quad c/2 + d \leq k_c(v_+), \quad c/2 - d \leq k_c(v_+), \\ v_+(e + c/2 + d) < R, \qquad v_+(c/2 - d) \leq R_B, \end{gathered} \tag{22}$$

along with $e \leq k_c(v_+)$ and $\ell(e + c) < k_j b$ and the common parameter bounds. This is not permission to use arbitrary imbalance at any load.

For an exact existence example, choose arbitrary positive speed scale u, density scale κ, lengths ℓ, h, and acceleration a_r. Put

$$\begin{gathered} v_- = u, \ v_+ = 2u, \ w = 2u, \ k_j = 3\kappa, \ \gamma = 2, \quad R = 39\kappa u/20, \quad R_B = \kappa u, \\ L_p = 2\ell, \quad b = \ell, \quad c_0 = 11/10, \quad \beta = 1/100, \quad e_0 = 3\kappa/5, \quad c = 11\kappa/20, \\ a_0 = 11\kappa/40 + d, \quad b_0 = 11\kappa/40 - d, \qquad |d| \leq 3\kappa/40. \end{gathered} \tag{23}$$

Then $r \in [4\kappa/5, 19\kappa/20]$, $M = 23\kappa\ell/20$, $k_c(v_+) = 3\kappa/2$ and $C(v_-) = 2\kappa u$. The worst ordinary E/A margin is $R - v_+ r \geq \kappa u/20 > 0$; the far margin is $R_B - v_+ b_0 \geq 3\kappa u/10 > 0$; $M/(k_j b) = 23/60 < 1$; and $\eta = 27/25 > 0$. All initial lanes have positive free-flow density with substantial margins.

At the two endpoints the A/B loads are swapped. At $v = v_+$ and the common initial preparation progress, the two alternatives in (5) are:

$(a_0, b_0)/\kappa$	Donor offer ve_0	Receiving alternative	Selected f
$(1/5, 7/20)$	$6\kappa u/5$	$13\kappa u/10$	$6\kappa u/5$
$(7/20, 1/5)$	$6\kappa u/5$	$23\kappa u/20$	$23\kappa u/20$

The first preparation is donor-limited and the second receiving-limited. The inequalities are strict, so this rate contrast persists over a positive initial progress interval by continuity. The actual density source is f/h. This is exact local algebra, with no traffic rollout or calibrated parameter recommendation. It establishes a genuine mechanistic distinction inside the admitted family, not indistinguishable histories, different optimal advice or useful gradients.

6 What transfers, what stops here, and provenance

The whole-forward result, complete mass accounting, own-lane-first preparation, monotone pulse, storage protection, roots, recent-mass recovery and TST transfer as stated above. The old initial-E-only reconstruction does not: E and M fix $a_0 + b_0$, not a_0. Symmetric matched-history and action examples remain valid within their old member but cannot certify the new Level 2-to-Level 3 target. AS2 must examine every permitted observation and history, including lawful inference. AS3 must separately qualify action relevance and the complete advice derivative. Neither an interior optimum nor a positive information advantage is guaranteed.

Reused derivations. This is an original dependency audit and initialization extension of the accepted review model. The supporting lineage is the feedback-8 consolidated mathematics, Sections 2–11 (initialization through global forward/TST), and the W-F whole-domain qualification. The precise source locations are retained separately in the AS1 component and proof dependency map. No new paper was viewed or new literature-backed mechanism selected for AS1. Existing literature supports the ingredients to the extent recorded in its source audits; it is not attributed this complete asymmetric law or the illustrative parameter values. Applied MP6 remains open. SUMO is optional. This document ends at the AS1 contract-confirmation checkpoint.